

Katherine Avanesov

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Education

California Institute of Technology (Pasadena, CA September 2023 - June 2027)

- Physics (Major) and Computer Science (Minor).
- Cumulative GPA: 4.0/4.0
- Relevant coursework: Operating Systems, Software Engineering, Database Systems, Machine Learning, Computational Physics, Quantum Computation, Mathematical Chaos, Mathematical Methods in Physics.

Experience

Quantitative Research Intern – Clear Street (Summer 2026).

- Deployed a Python risk engine to production, integrating with Snowflake database via SQL for daily recalibration with a front-end dashboard providing risk managers with per-position cost breakdowns.
- Developed an option volatility-risk model spanning a 6,000+ stock universe to estimate liquidation cost and set margin requirements for institutional portfolios.
- Designed an algorithm that recognizes offsetting risks across a portfolio, reducing over-conservative margin estimates by 20%+ for diversified positions by decomposing implied-volatility fluctuations into normal modes of a multidimensional risk surface.

Researcher – CERN / Princeton CMS Group (Summers '24 – '25).

- 10 weeks on-site at CERN in Geneva, Switzerland under short-term User contract.
- Led the development of neural network-based classifiers for particle detection. Improved signal selection by over two orders of magnitude. Presented at American Physical Society Global Physics Summit 2025: Search for Long-Lived Particles with Hadronic Calorimeter Depth Segmentation.

Teaching Assistant – Caltech (Winter, Spring 2026):

- Supported 100+ students in Caltech's foundational courses CS2 (Introduction to Programming Methods, Java) and CS3 (Introduction to Software Design, C) through weekly office hours, providing conceptual and debugging help.

Projects

“NanoDB”: SQL Database Engine (Winter 2026).

- Built core components of a relational database across a 75,000+ line Java codebase spanning 15 subsystems, including low-level file storage, B+ tree indexes, and a query execution pipeline.
- Implemented external-memory algorithms, table-statistics tracking, and a dynamic-programming query optimizer that used those statistics to estimate execution costs and select efficient execution plans.

Java Virtual Machine (Fall 2025).

- Implemented a minimal Java Virtual Machine supporting class-file parsing, bytecode execution, and operand stack management. Validated the interpreter on sorting algorithms, recursive programs, and larger workloads including Collatz trajectories for 100k integers.

Spreadsheet Engine (Winter 2026).

- Collaborated with a partner over 10 weeks to build a spreadsheet engine supporting formula evaluation, function invocation, dependency tracking, and circular-reference detection.
- Improved runtime performance by up to 10x by optimizing the underlying cell-dependency graph with an iterative topological sort algorithm and batched cell-update logic.

Skills

- Languages: Python, C, Java, SQL. Fluent Spanish, French, Russian. Elementary Japanese.
- Tools: Git, Linux, Pandas, TensorFlow, PyTorch

Awards

- 2026 The George W. and Bernice E. Green Memorial Prize nominee for creative scholarship beyond coursework
- 2025 Caltech Margie Lauritsen Leighton Prize for academic excellence in Physics
- 2025 Caltech Hallet Smith Prize for best essay on Literature
- 2025 Bill Gross Business Plan Competition Finalist
- 2025 Taylor Lawrence SURF Research Fellow
- 2024 Caltech Robert F. Christy Prize for an Outstanding Freshman in Physics
- 2024 Samuel P. and Frances Krown SURF Research Fellow
- 2023 Top Gold Round 1 and Silver Round 2 British Physics Olympiad; Gold British Chemistry Olympiad